

IF 45 I

Advance Web Programming

Meet7 - CRUD Operations in MySQL Database

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Meet 7

Objective:

Sub-CLO07I3 - Students are able to implement various types of libraries and database to create web application using Node JS (C3)

OUTLINE

- *Find, findOne, insertOne, updateOne, deleteOne*

Find, findOne, insertOne, updateOne, deleteOne

Find, findOne, insertOne, updateOne, deleteOne

find() pada Mysql adalah: (**R**ead)

- `SELECT * FROM ms_mahasiswa;`

findOne() pada Mysql adalah: (**R**ead)

- `SELECT * FROM ms_mahasiswa WHERE nim='12345' LIMIT 1;`

insertOne() pada Mysql adalah: (**C**reate)

- `INSERT INTO ms_mahasiswa (...) VALUES (...);`

updateOne() pada Mysql adalah: (**U**ppdate)

- `UPDATE ms_mahasiswa SET nama='Budi' WHERE nim='12345';`

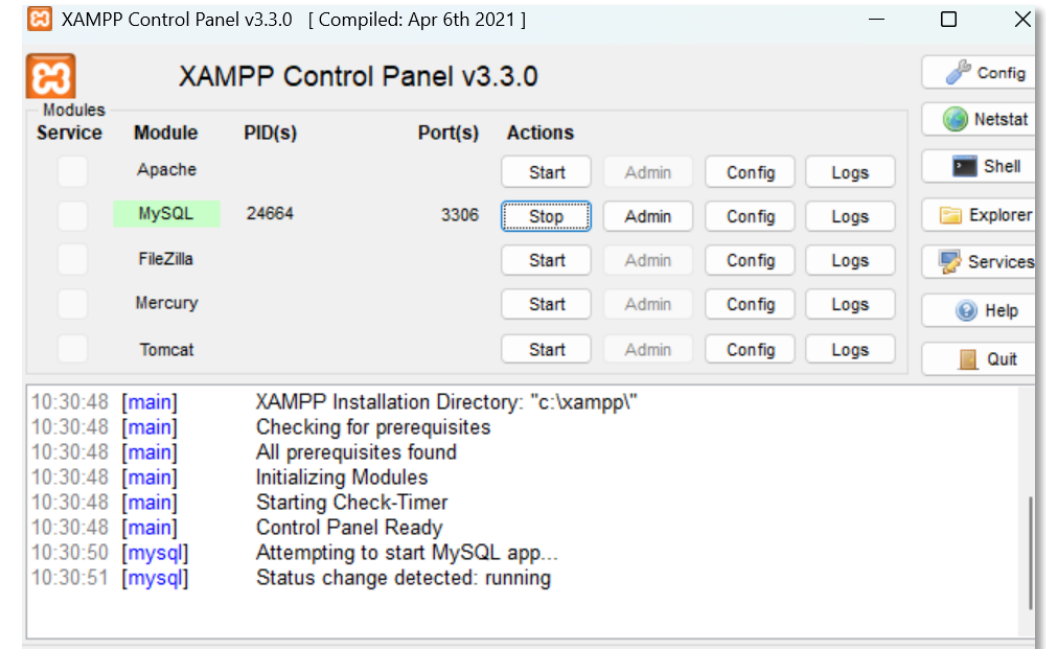
deleteOne() pada Mysql adalah: (**D**elete)

- `DELETE FROM ms_mahasiswa WHERE nim='12345';`

Preparing Database and Module

Like the previous meeting, to access database via NodeJS, you need to:

1. Install XAMPP and then run mysql
2. Make sure, you have database with database name is “dbkampus” and table, with name “ms_mahasiswa”
3. Install module via terminal, with command “npm install mysql2 readline-sync”



```
t\meet7> npm install mysql2 readline-sync
```

Preparing Database Connection

Beside is the sample code to connect with your database mysql.

Don't forget to save your file with name "koneksi.js"

```
const mysql = require("mysql2");

const connection =
mysql.createConnection({host:"localhost",user:"
root",password:"",database:"dbkampus"});

connection.connect(err => {
    if(err) {
        console.log("ada error " + err);
    } else {
        console.log("Koneksi berhasil");
    }
});

module.exports = connection;
```

Create Load Data files - Read (I)

Create new file, with name “tampil_data.js” and write the code beside.

```
const kon = require("../koneksi.js");

const load_data_mhs = (res) => {
  const q = "SELECT * FROM ms_mahasiswa";
  const sql = kon.query(q, (err, result) => {
    if(err) {
      console.log("error query " + err);
    } else {
      res.write("<table border='1' rules='all' >");
      res.write("<tr bgcolor='yellow' >");
      res.write("<td>Nim</td>");
      res.write("<td>Nama</td>");
      res.write("<td>Jurusan</td>");
      res.write("<td>Alamat</td>");
      res.write("<td>Telepon</td>");
      res.write("</tr>");
    }
  });
}
```


Create Load Data files - Read (2)

Create new file, with name “tampil_data.js” and write the code beside.

```
        result.forEach((elm)=>{
            res.write("<tr>");
            res.write("<td>" + elm.nim + "</td>");
            res.write("<td>" + elm.nama + "</td>");
            res.write("<td>" + elm.jurusan + "</td>");
            res.write("<td>" + elm.alamat + "</td>");
            res.write("<td>" + elm.telepon + "</td>");
            res.write("</tr>");
        });
        res.write("</table>")
    }
    res.end();
});
}

module.exports = load_data_mhs;
```

Call other files

Create your main file, with name “app.js” and write the sample code beside to include your tampil_data.js files

```
const dataMhs = require("../tampil_data");
const http = require("http");
const fs = require("fs");

const server = http.createServer((req, res) => {
  res.writeHead(200, {"content-type": "text/html"});

  fs.readFile("form_input.html", (err, data) => {
    if(err) {
      console.log("Gagal panggil form input");
      res.end();
    } else {
      res.write(data);
      dataMhs(res);
    }
  });
});

server.listen(3000, () => {
  console.log("server berjalan di http://localhost:3000");
})
```

Result

← → ↻ ⓘ localhost:3000

Nim:

Nama:

Jurusan:

Alamat:

Telepon:

simpan

Nim	Nama	Jurusan	Alamat	Telepon
111	Wirawan	Informatika	Gading Serpong	0123456
222	Budi	Sistem Informasi	Jalan Scientia	02231
3456	Yanto	DKV	Dimana aja	12323123

Save data – Create (I)

To save, create new file with name “simpan_data.js” and write the sample code beside

```
const querystring = require("querystring");
const kon = require("../koneksi.js");

const simpan_data_mhs = (req, res) => {
  let isiData = "";
  req.on('data', chunk => {
    isiData += chunk;
  });
  req.on('end', () => {
    const formData = querystring.parse(isiData);
    console.log("formData: " + JSON.stringify(formData));

    const sql = "INSERT INTO ms_mahasiswa (nim, nama, jurusan, alamat, telepon) VALUES (?, ?, ?, ?, ?)";
    kon.query(sql, [formData.txtNim, formData.txtNama, formData.txtJurusan, formData.txtAlamat, formData.txtTelp], (err, result) => {
    {
```

Save data – Create (2)

To save, create new file with name “simpan_data.js” and write the sample code beside

```
        if (err) {
            console.error("Gagal insert:", err);
            res.end();
            return;
        } else {
            console.log("Simpan Berhasil");
            res.writeHead(302, { 'Location': 'http://localhost:3000' });
            res.end();
        }
    });
});
}

module.exports = simpan_data_mhs;
```

Editing app.js to call simpan_data.js files

Try to edit your app.js to call “simpan_data.js” files.

```
const dataMhs = require("../tampil_data");
const http = require("http");
const fs = require("fs");
const simpanMhs = require("../simpan_data");

const server = http.createServer((req, res) => {
  res.writeHead(200, {"content-type": "text/html"});

  console.log(req.method + " / " + req.url);
  if(req.method=="POST" && req.url=="/simpan") {
    simpanMhs(req, res);
  } else {
    fs.readFile("form_input.html", (err, data)=> {
```

Editing User Interface for Delete and Edit button

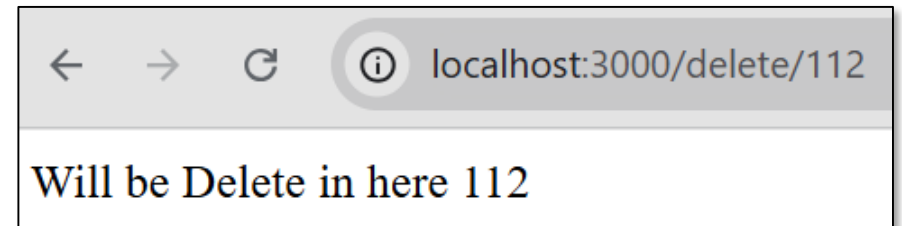
Try to edit “tampil_data.js” files, and add this sample code below

```
result.forEach((elm)=>{
    res.write("<tr>");
    res.write("<td>" + elm.nim + "</td>");
    res.write("<td>" + elm.nama + "</td>");
    res.write("<td>" + elm.jurusan + "</td>");
    res.write("<td>" + elm.alamat + "</td>");
    res.write("<td>" + elm.telepon + "</td>");
    res.write("<td align='center' ><a href='/delete/'+"+elm.nim+"'> x </a></td>");
    res.write("<td align='center' ><a href='/edit/'+"+elm.nim+"'> edit </a></td>");
    res.write("</tr>");
});
```

Nim	Nama	Jurusan	Alamat	Telepon	Delete	Edit
111	Wirawan	Informatika	Gading Serpong	0123456	<u>x</u>	<u>edit</u>
112	Ibu	Informatika	kjhasjkdh akdhk	12123	<u>x</u>	<u>edit</u>

Routing to Delete and Edit

```
const server = http.createServer((req, res) => {  
  
  const urlsplit = req.url.split("/");  
  
  if(req.method=="POST" && urlsplit[1]=="simpan") {  
    simpanMhs(req, res);  
  } else if (req.method=="GET" && urlsplit[1]=="edit") {  
    res.writeHead(200, {"content-type":"text/html"});  
    res.write("Will be Edit in here " + urlsplit[2]);  
    res.end();  
  } else if (req.method=="GET" && urlsplit[1]=="delete") {  
    res.writeHead(200, {"content-type":"text/html"});  
    res.write("Will be Delete in here " + urlsplit[2]);  
    res.end();  
  } else {  
    res.writeHead(200, {"content-type":"text/html"});  
  
    fs.readFile("form_input.html", (err, data)=> {
```



Delete files

Create new file with name “delete_data.js” and write the sample code beside

```
const kon = require("../koneksi.js");

const hapus_data_mhs = (req, res, nim) => {
  console.log("mahasiswa yang akan dihapus: " + nim);

  const sql = "DELETE FROM ms_mahasiswa WHERE nim = ?";
  kon.query(sql, [nim], (err, result) => {
    if (err) {
      console.error("Gagal Hapus:", err);
      res.end();
      return;
    } else {
      console.log("Hapus Berhasil");
      res.writeHead(302, { 'Location': 'http://localhost:3000' });
      res.end();
    }
  });
});

module.exports = hapus_data_mhs;
```

Assessment

Make the Edit function work, When user try to edit, user can edit any value except NIM field.

Write your script and save zip your project folder and rename to **nim_nama_meet7.zip**

Edit app.js To call delete_data.js files

Try to edit app.js and write like the code beside.

```
const simpanMhs = require("../simpan_data");
const hapusMhs = require("../delete_data");

...
...
} else if (req.method=="GET" && urlsplit[1]=="edit") {
    res.writeHead(200, {"content-type":"text/html"});
    res.write("Will be Edit in here " + urlsplit[2]);
    req.end();
} else if (req.method=="GET" && urlsplit[1]=="delete") {
    hapusMhs(req, res, urlsplit[2]);
} else {
    res.writeHead(200, {"content-type":"text/html"});
```

NEXT WEEK'S OUTLINE

- Introduction to Express.js

REFERENCES

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- The Node.js Foundation, “Node.js Documentation,” [Online]. Available: <https://nodejs.org/en/docs/>. [Accessed: Jun. 5, 2025].
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Visi

Menjadi Program Studi Strata Satu Informatika **unggulan** yang menghasilkan lulusan **berwawasan internasional** yang **kompeten** di bidang Ilmu Komputer (*Computer Science*), **berjiwa wirausaha** dan **berbudi pekerti luhur**.



Misi

1. Menyelenggarakan pembelajaran dengan teknologi dan kurikulum terbaik serta didukung tenaga pengajar profesional.
2. Melaksanakan kegiatan penelitian di bidang Informatika untuk memajukan ilmu dan teknologi Informatika.
3. Melaksanakan kegiatan pengabdian kepada masyarakat berbasis ilmu dan teknologi Informatika dalam rangka mengamalkan ilmu dan teknologi Informatika.